A DUALIZING COMPLEX ON A LOCALLY NOETHERIAN SCHEME INDUCES AN ANTI-EQUIVALENCE OF THE BOUNDED COHERENT DERIVED CATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A89 (sources/stacks-duality/pinned-original-source.tex line 472).

Allowed standard prerequisites (editorial).

- Comparison of the scheme level and affine level dualizing functors. Same chapter, Lemmas labelled lemma-equivalent-definitions and lemma-affine-duality, cited at proof lines 491-506. Shows the functor D is induced by the algebraic dualizing functor on affines.
- The algebraic dualizing functor is an anti-equivalence with id -> D o D. Dualizing Complexes, Lemma labelled dualizing-lemma-dualizing, cited at proof lines 518-522 and 524-526. Supplies the isomorphism and the boundedness statements.
- The evaluation map between internal Homs. Cohomology, Lemma labelled cohomology-lemma-internal-hom-evaluate, cited at proof lines 512-516. Provides the second arrow of the displayed composition.
- The evaluation isomorphism alternative. Same chapter, Lemma labelled lemma-internal-hom-evaluate-isom part (2), cited in the footnote at proof lines 520-524. Gives the alternative route to the isomorphism.

Source reference. Original lemma 0A86: sources/stacks-duality/pinned-original-source.tex, source lines 308–308. Same-chapter pointer into the author environment lemma at source line 308; invoked at proof line 493 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A88: sources/stacks-duality/pinned-original-source.tex, source lines 366–366. Same-chapter pointer into the author environment lemma at source line 366; invoked at proof line 494 of the counted proof. It is

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Source reference. Original lemma 0A8U: sources/stacks-chow/tags.txt, source lines 12927–12927. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 512 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A7C: sources/stacks-chow/tags.txt, source lines 12875–12875. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 515 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4I: sources/stacks-duality/pinned-original-source.tex, source lines 415–415. Same-chapter pointer into the author environment lemma at source line 415; invoked at proof line 519 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let K be a dualizing complex on a locally Noetherian scheme X. Then K is an object of $D_{Coh}(\mathcal{O}_X)$ and $D = R\mathcal{H}om_{\mathcal{O}_X}(-, K)$ induces an anti-equivalence*

$$D : D_{Coh}(\mathcal{O}_X) \longrightarrow D_{Coh}(\mathcal{O}_X)$$

which comes equipped with a canonical isomorphism $id \to D \circ D$. If X is quasi-compact, then D exchanges $D^+_{Coh}(\mathcal{O}_X)$ and $D^-_{Coh}(\mathcal{O}_X)$ and induces an anti-equivalence $D^b_{Coh}(\mathcal{O}_X) \to D^b_{Coh}(\mathcal{O}_X)$.

Proof. Let $U \subset X$ be an affine open. Say $U = \mathrm{Spec}(A)$ and let $\omega_A^\bullet$ be a dualizing complex for A corresponding to $K|_U$ as in Lemma 0A86. By Lemma 0A88 the diagram

$$\begin{array}{ccc} D_{Coh}(A) & \longrightarrow & D_{Coh}(\mathcal{O}_U) \\ {\scriptstyle R\mathrm{Hom}_A(-,\omega_A^\bullet)}\downarrow & & \downarrow{\scriptstyle R\mathcal{H}om_{\mathcal{O}_X}(-,K|_U)} \\ D_{Coh}(A) & \longrightarrow & D(\mathcal{O}_U) \end{array}$$

commutes. We conclude that D sends $D_{Coh}(\mathcal{O}_X)$ into $D_{Coh}(\mathcal{O}_X)$. Moreover, the canonical map

$$L \longrightarrow R\mathcal{H}om_{\mathcal{O}_X}(K, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X} L \longrightarrow R\mathcal{H}om_{\mathcal{O}_X}(R\mathcal{H}om_{\mathcal{O}_X}(L, K), K)$$

(using Cohomology, Lemma 0A8U for the second arrow) is an isomorphism for all L because this is true on affines by Dualizing Complexes, Lemma 0A7C[1] and we have already seen on affines that we recover what happens in algebra. The statement on boundedness properties of the functor D in the quasi-compact case also follows from the corresponding statements of Dualizing Complexes, Lemma 0A7C. □

[1] An alternative is to first show that $R\mathcal{H}om_{\mathcal{O}_X}(K, K) = \mathcal{O}_X$ by working affine locally and then use Lemma 0G4I part (2) to see the map is an isomorphism.